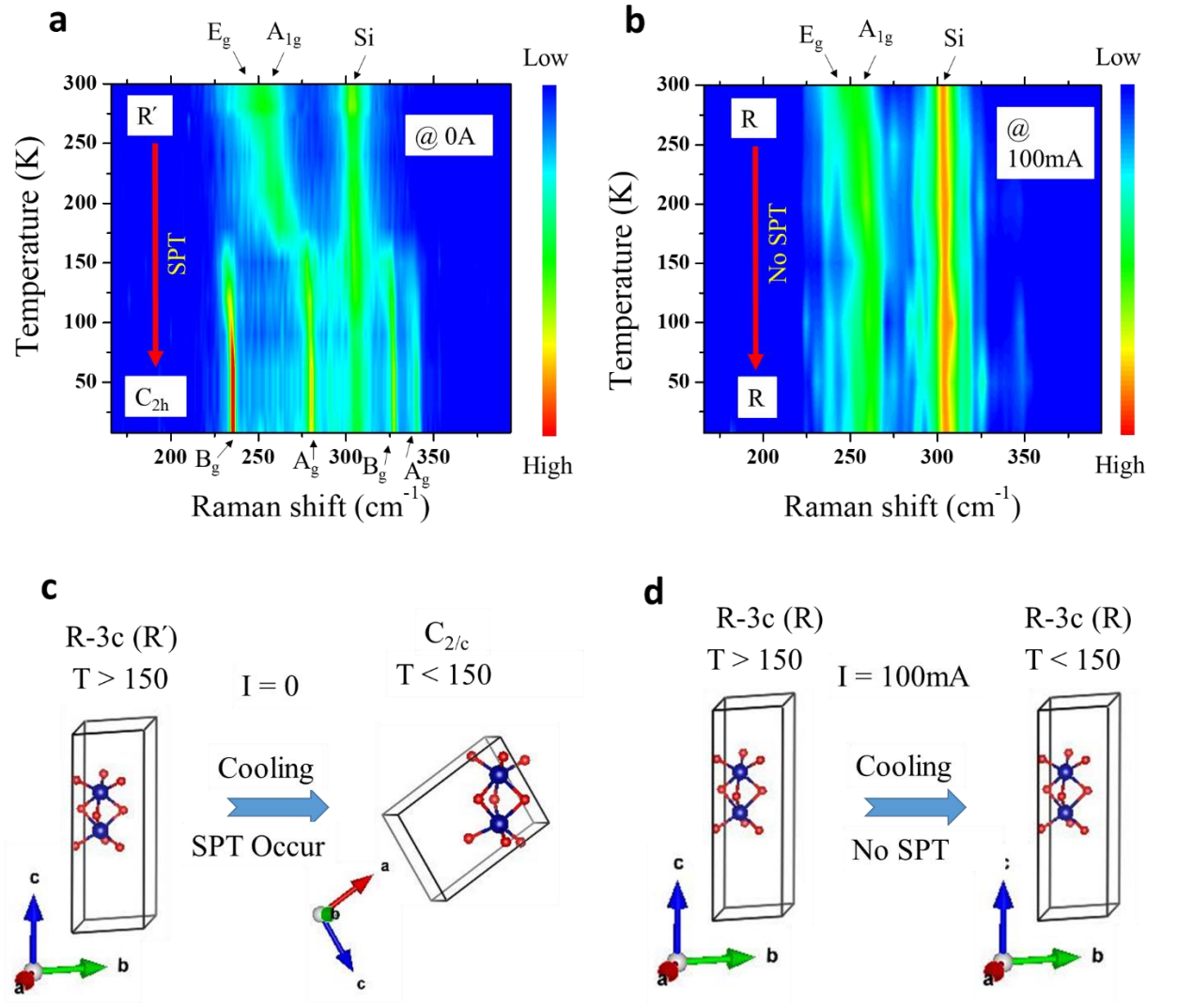

Fig. 4| Temperature evolution of the microstructure under zero and finite electric field. a, Contour plot of Raman spectra collected on V_2O_3/Si under zero current and **b,** under 100 mA dc current. **c & d,** Temperature evolution of the crystal structure measured under 0A and 100mA current cooled mode. Cartoon depicts R-3c crystal structural transformation to $C_{2/c}$ structure below 150K under zero current, and this transition is suppressed under 100mA current.

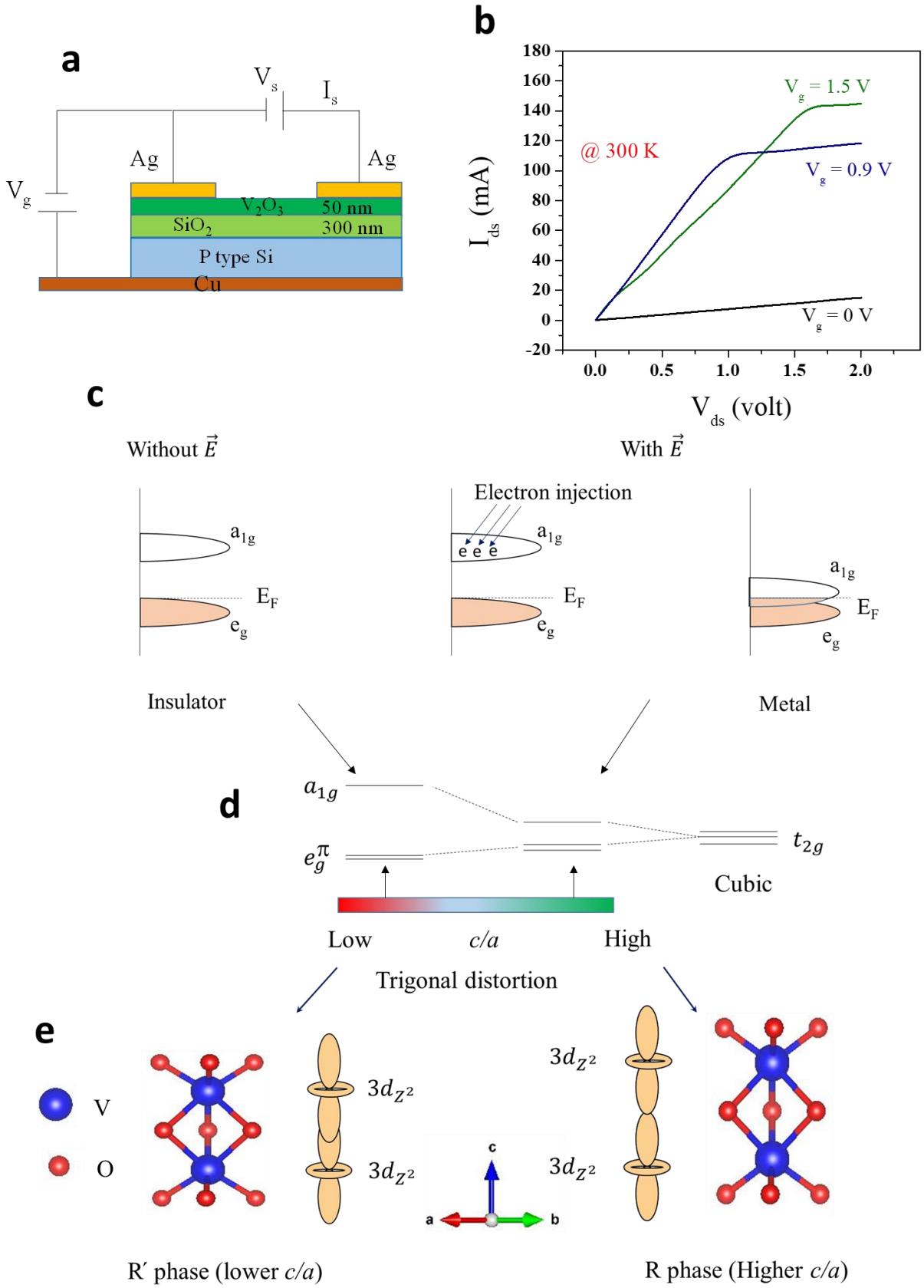
a
V_s
I_s
Ag
Ag
V_g
V_2O_3
50 nm
SiO_2
300 nm
P type Si
Cu
b
@ 300 K
V_g = 1.5 V
V_g = 0.9 V
V_g = 0 V
I_{ds} (mA)
V_{ds} (volt)
c
Without $\vec{E}$
With $\vec{E}$
Electron injection
a_{1g}
E_F
e_g
Insulator
Metal
d
a_{1g}
e_g^{π}
t_{2g}
Cubic
Low
c/a
High
Trigonal distortion
e
V
O
$3d_{Z^2}$
$3d_{Z^2}$
$3d_{Z^2}$
$3d_{Z^2}$
c
a
b
R′ phase (lower c/a)
R phase (Higher c/a)

Fig. 5| Mott field effect transistor characteristics and Mechanism of the electric field induced structural change in V_2O_3/Si thin film. a, Schematic of Mott field effect transistor (MFET) structure. **b,** The channel I-V characteristics at room temperature under different gate voltages. **c,** Schematics of the energy band diagram in insulating and metallic state. **d,** Electronic band splitting under the trigonal distortion with different *c/a* ratio values. **e,** Two face sharing VO_6 octahedra corresponding to R′ and R phase.

Materials and methods

Growth and device fabrication: The V_2O_3 thin film was deposited on 285nm SiO_2 coated (100) P type Si substrate using Pulse Laser deposition method (PLD). A sintered pellet of V_2O_5 was used as a target and 248nm KrF pulse Eximer laser of pulse energy 2J/cm^2 and repetition rate 3Hz was used for deposition. Substrate was kept at a distance of 5cm from the target. During deposition substrate temperature was 650°C and the deposition chamber was evacuated to 10^{-6} mbar. After deposition the film was cooled down to room temperature with a rate of 2.2 K/min while maintaining the same pressure. 30 nm Au electrodes were deposited on V_2O_3 thin film using magneton sputtering method for four probe resistivity measurements. For bottom gated three terminal device, we stacked the P type Si with Cu using Silver paste. The device structure is presented in Fig. 5a.

Structural characterization: The XRD θ-2θ and X ray reflectivity (XRR) measurements were performed using PAN analytical X'PERT high resolution x-ray diffraction (HRXRD) system equipped with a Cu anode. Raman measurements were performed using Horiba Jobin-Yvon, France, micro Raman spectrometer equipped with a 633 nm excitation laser, 1800 g/mm grating, an edge filter for Rayleigh line rejection and a CCD detector giving a spectral resolution of ~1cm^{-1}. The Laser was focused to a spot size of ~1 micron onto the surface of the sample using a 50× objective lens. The spectra were collected in back scattering geometry. A liquid helium flow cryostat from Janis (USA) was used for cooling the sample temperature. In situ resistivity and Raman measurements at different T were performed using this cryostat.

TEM: STEM images were collected using aberration (probe) corrected Titan Themis 300 operated at 300kV equipped with SuperXG quad EDS detector. 24 mrad convergence angle used for EDS and HAADF collection angle of 48-200 mrad were used at a camera length of 160mm.

Resistivity measurements: Resistivity and I-V characteristics were performed in the linear four point probe configuration using Keithley 2425 source meter and Keithley 2002 multimeter. Temperature was controlled using the Lakeshore 335 temperature controller. Temperature dependent resistivity measurements in cooling and heating cycles were performed in the temperature sweeping mode with 3K/min temperature sweeping rate. The resistance measurements under different current were performed by applying some specific constant dc current and the corresponding voltage drop was measured. Isothermal I-V characteristics were obtained in continuous increment/decrement of current and measuring the corresponding voltage difference between two probing points. For measuring the isotherm at a particular temperature, each time the sample was cooled from 300K with a cooling rate 5K/minute. After reaching the desired temperature, isotherm was obtained using the sequence 0A→0.1A→0A→-0.1A→0A. The sequence of minor hysteresis measurement is shown graphically in Fig. S2. Electrodes separation was 550μm.

Acknowledgement